Ansys
2026 R1

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Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	3d, pressure-based, laminar
Version	26.1.0-10262
Source Revision	b6f454dde8
Build Time	Feb 2 2026 16:24:29 EST
CPU	AMD Ryzen 5 5600H with Radeon Graphics
OS	Windows

Geometry and Mesh

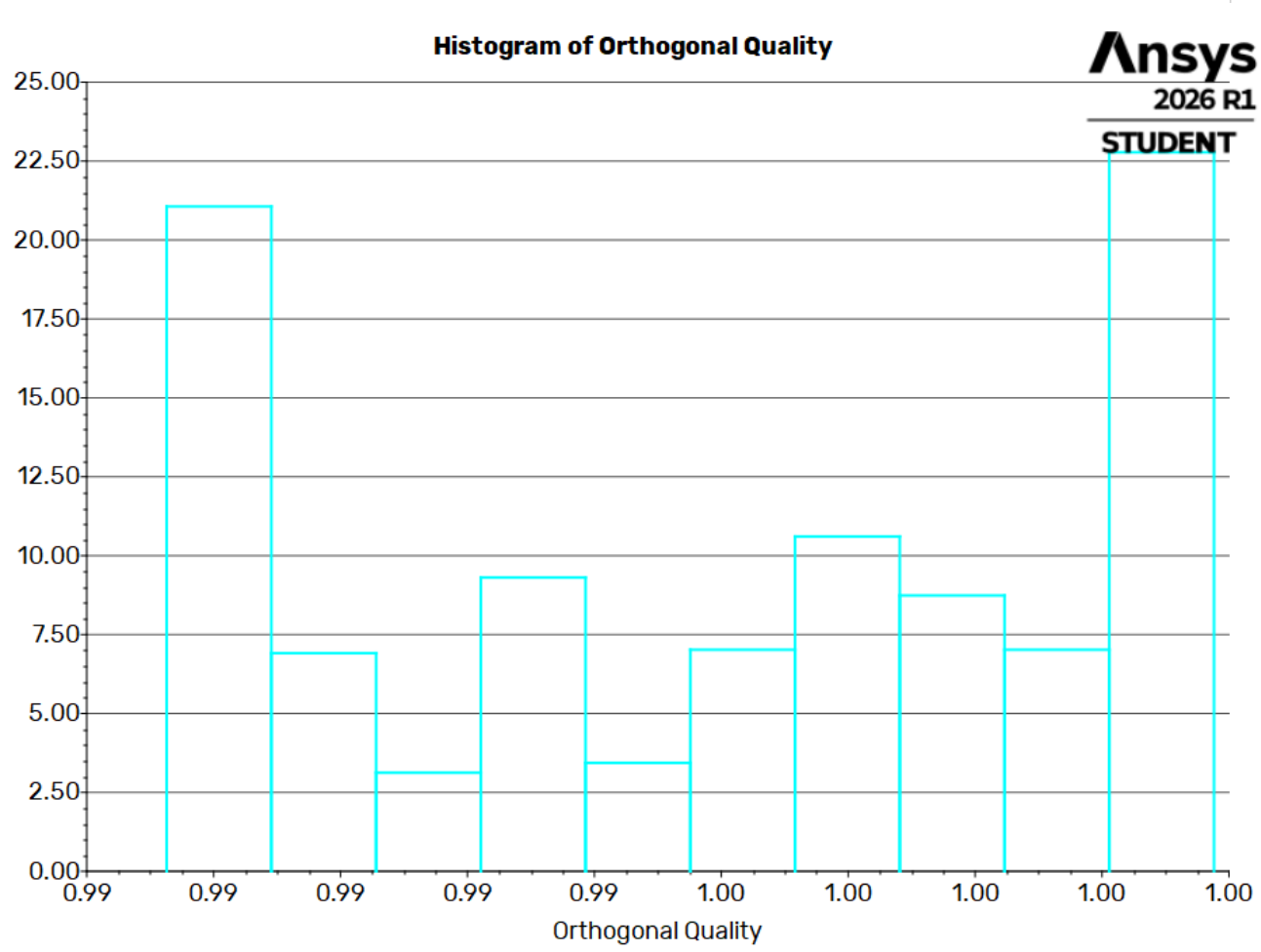
Mesh Size

Cells	Faces	Nodes
8960	28304	10455

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
solid	Hex Cell	0.99162966	4.9696031

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	Laminar
Heat Transfer	Enabled
Radiation	Surface to Surface

Material Properties

– Fluid	
– air	
Density	incompressible ideal gas
Cp (Specific Heat)	1021 J/(kg K)
Thermal Conductivity	0.0371 W/(m K)
Viscosity	2.485e-05 kg/(m s)
Molecular Weight	28.966 kg/kmol
– Solid	
– insulation	
Density	50 kg/m^3
Cp (Specific Heat)	800 J/(kg K)
Thermal Conductivity	0.09 W/(m K)

Cell Zone Conditions

– Fluid	
– solid	

Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Porous zone?	no
3D Fan Zone?	no

Boundary Conditions

– Wall	
– w-high-x	
Wall Thickness [m]	0.05
Heat Generation Rate [W/m^3]	0
Material Name	insulation
Thermal BC Type	Mixed
Convective Heat Transfer Coefficient [W/(m^2 K)]	5
Free Stream Temperature [K]	293.15
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Internal Emissivity	1
External Emissivity	0.75
External Radiation Temperature [K]	293.15
Radiation BC Type	Opaque
Participates in View Factor Calculation?	yes
Faces Per Surface Cluster	1
Convective Augmentation Factor	1
– w-high-y	
Wall Thickness [m]	0.95
Heat Generation Rate [W/m^3]	0.05
Material Name	insulation
Thermal BC Type	Mixed

Convective Heat Transfer Coefficient [W/(m^2 K)]	3
Free Stream Temperature [K]	293.15
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Internal Emissivity	1
External Emissivity	0.75
External Radiation Temperature [K]	293.15
Radiation BC Type	Opaque
Participates in View Factor Calculation?	yes
Faces Per Surface Cluster	1
Convective Augmentation Factor	1
– w-low-y	
Wall Thickness [m]	0.95
Heat Generation Rate [W/m^3]	0.05
Material Name	insulation
Thermal BC Type	Mixed
Convective Heat Transfer Coefficient [W/(m^2 K)]	3
Free Stream Temperature [K]	293.15
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Internal Emissivity	1
External Emissivity	0.75
External Radiation Temperature [K]	293.15
Radiation BC Type	Opaque
Participates in View Factor Calculation?	yes
Faces Per Surface Cluster	1
Convective Augmentation Factor	1
– w-low-x	

Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	insulation
Thermal BC Type	Temperature
Temperature [K]	473.15
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Internal Emissivity	0.95
Radiation BC Type	Opaque
Participates in View Factor Calculation?	yes
Faces Per Surface Cluster	1
Convective Augmentation Factor	1
– w-high-z	
Wall Thickness [m]	0.05
Heat Generation Rate [W/m^3]	0
Material Name	insulation
Thermal BC Type	Mixed
Convective Heat Transfer Coefficient [W/(m^2 K)]	5
Free Stream Temperature [K]	293.15
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Internal Emissivity	1
External Emissivity	0.75
External Radiation Temperature [K]	293.15
Radiation BC Type	Opaque
Participates in View Factor Calculation?	yes
Faces Per Surface Cluster	1
Convective Augmentation Factor	1

– w-low-z	
Wall Thickness [m]	0.05
Heat Generation Rate [W/m^3]	0
Material Name	insulation
Thermal BC Type	Mixed
Convective Heat Transfer Coefficient [W/(m^2 K)]	5
Free Stream Temperature [K]	293.15
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Internal Emissivity	1
External Emissivity	0.75
External Radiation Temperature [K]	293.15
Radiation BC Type	Opaque
Participates in View Factor Calculation?	yes
Faces Per Surface Cluster	1
Convective Augmentation Factor	1

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	solid

Solver Settings

– Equations	
Flow	True
Energy	True
– Numerics	
Absolute Velocity Formulation	True
– Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Energy	0.75
Explicit Momentum	0.5
Explicit Pressure	0.5
– Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
– Discretization Scheme	
Pressure	Body Force Weighted
Momentum	Second Order Upwind
Energy	Second Order Upwind
– Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

Number of Machines	1
Number of Cores	1
Case Read	13.632 seconds
Data Read	0.428 seconds
Iteration	53.632 seconds
AMG	24.487 seconds
Virtual Current Memory	2.93004 GB
Virtual Peak Memory	3.04763 GB
Memory Per M Cell	252.458

Solution Status

Iterations: 600

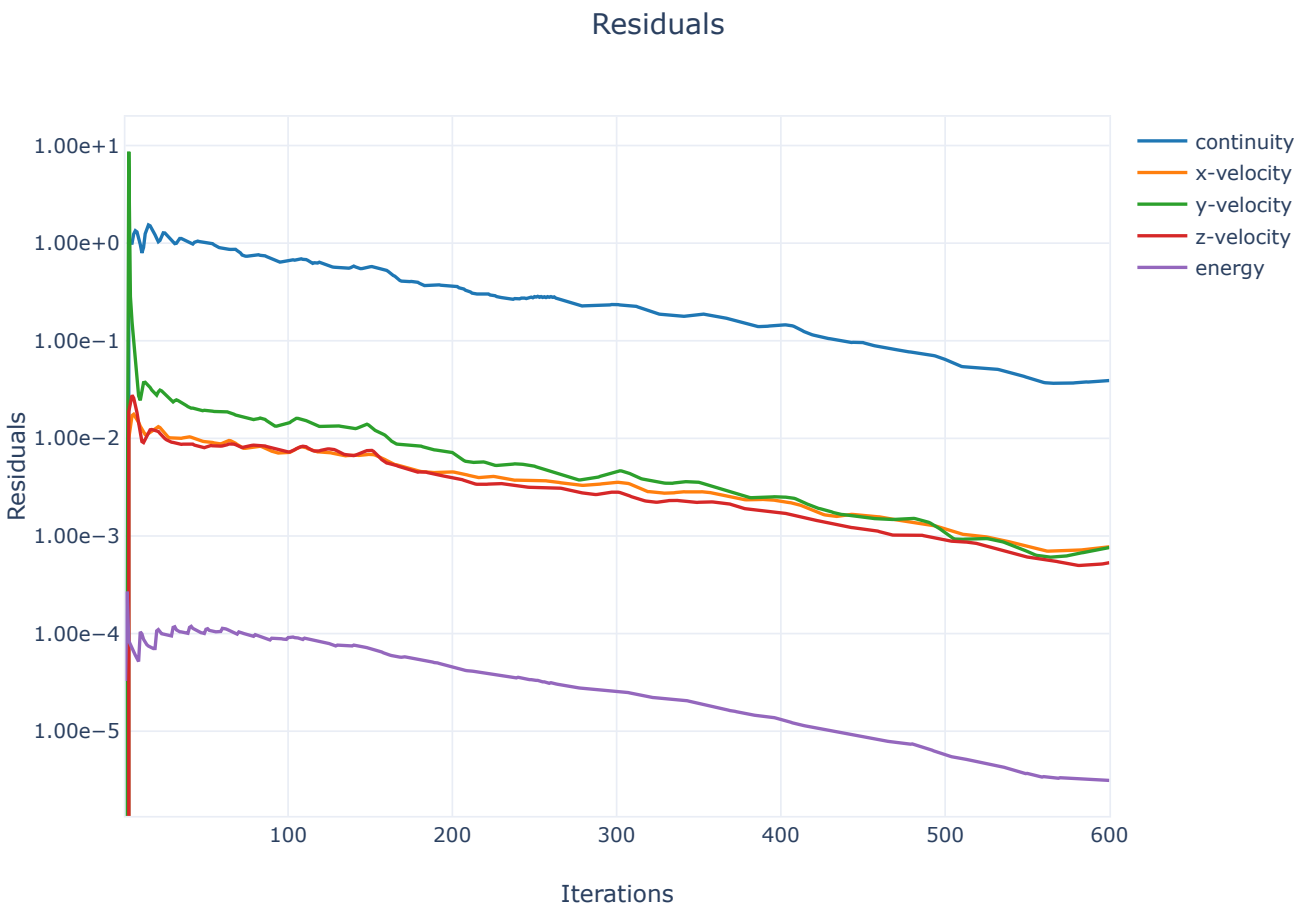
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continuity	0.03913837	0.001	Not Converged
x-velocity	0.0007750211	0.001	Converged
y-velocity	0.0007616651	0.001	Converged
z-velocity	0.0005327863	0.001	Converged
energy	3.132311e-06	1e-06	Not Converged

Report Definitions

parameter-2	-0.8811617	W
parameter-1	-2.445026	W

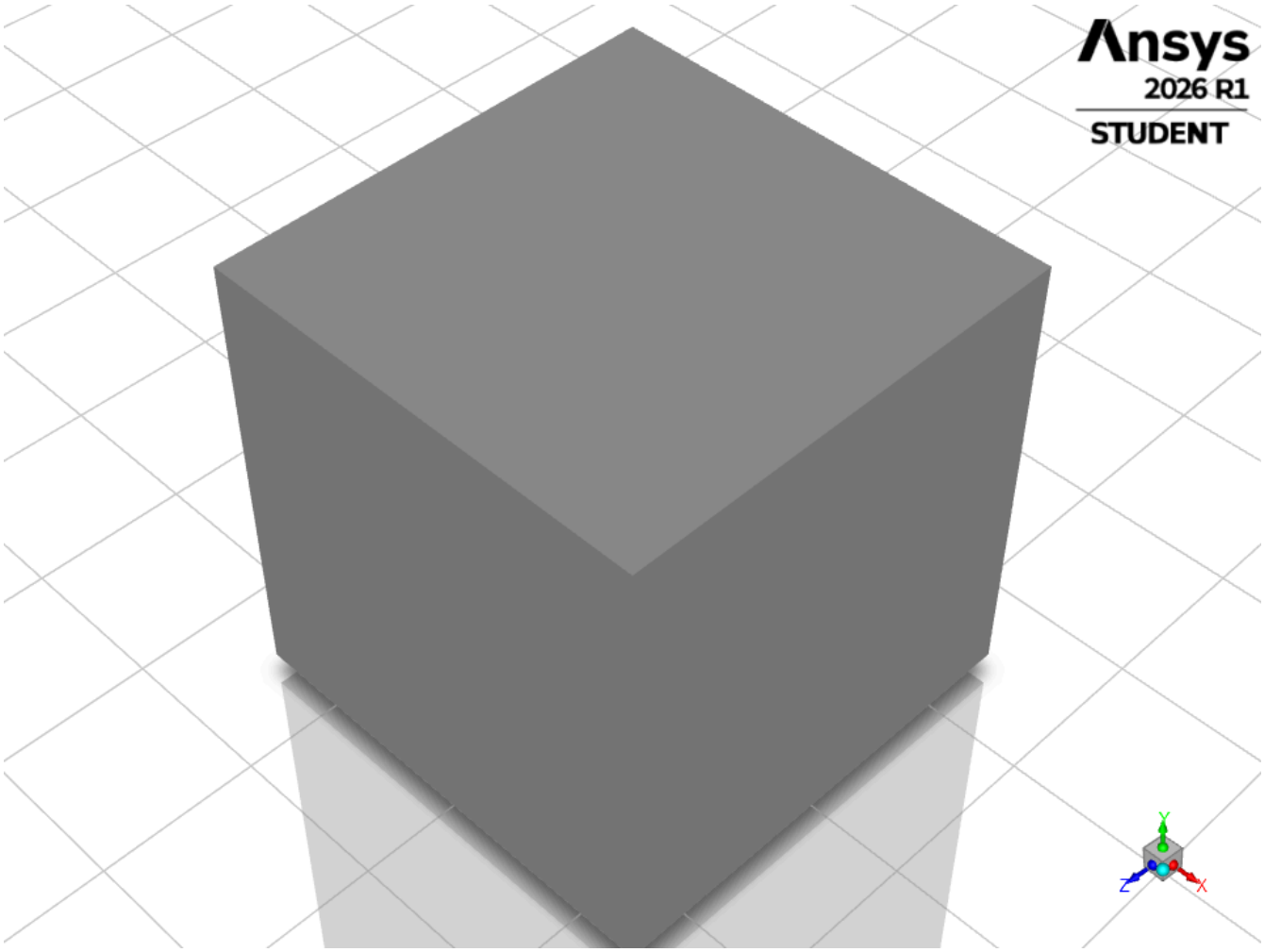
Plots

Residuals



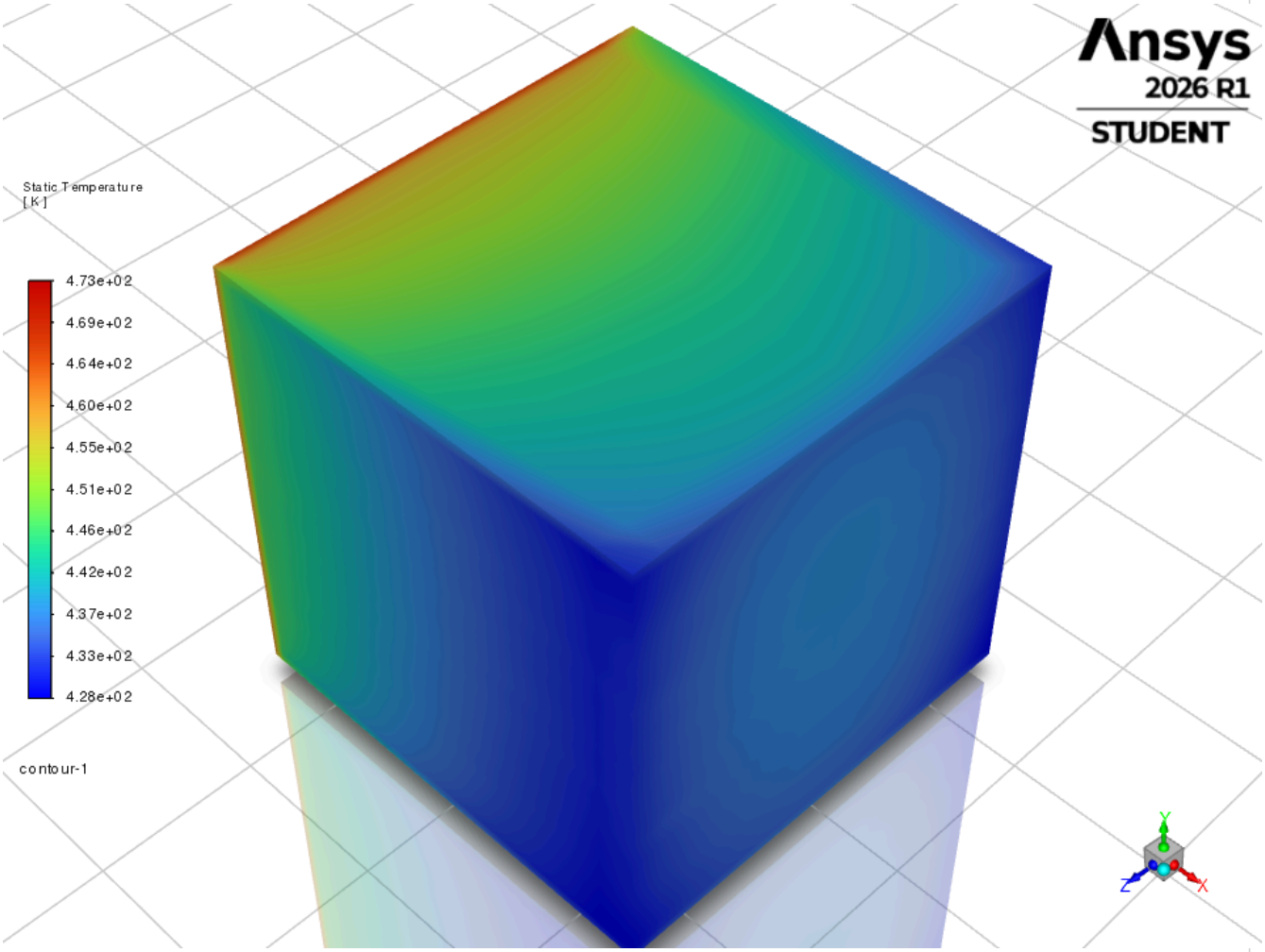
Mesh

mesh-1


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STUDENT
Y
Z
X

Contours

contour-1



Vectors

vector-1

